GENETIC VARIANCES IN SUB-POPULATIONS USING RELATIONSHIP MATRICES

In lecture 7 and in the examples I talk about genetic variances and what they mean. A variance is a property of a population and is estimated in software such as ASReml, and it is a relevant question to ask: what population does that variance refer to? For example when studies analyse the same data with a pedigree model and a genomic model, and then use the genetic variances to calculate the heritabilities in the standard way, are those heritabilities estimating the same thing or something different – they are often compared as if they should be the same.

NUMERATOR RELATIONSHIP MATRIX, A

The numerator relationship matrix uses the concept of the base generation which is a group of unrelated individuals with no IBD between any pair of alleles. In discussing what happens to say SNP alleles in the base we have been vague on what is meant by allele frequency in the base when talking about IBS. If we say the frequency of a particular allele (say the double muscling mutation) is p in the base population then p is the frequency of that allele in an imaginary infinite pool of alleles. Therefore the frequency of an individual being a homozygote in the base generation is the probability of two alleles taken at random from this pool being IBS, which is p^2, i.e. binomial probabilities.

However if I look at the s individuals in my base generation and their $2s$ alleles, the distribution of my allele is Binomial($2s$,p). The mean is $\bar{p}$ which need not be exactly the same as p, as the $2s$ alleles are only a sample from this imaginary infinite pool. Whilst E[$\bar{p}$] = p there will be a variance which is the variance of an estimate of p in a binomial distribution, i.e. $p(1-p)/2s$. This becomes small as s becomes large. The variance arises from drawing a sample from the imaginary infinite pool.

Turning now to breeding values: in forming the imaginary infinite pool from which our actual base population is sampled, the breeding values have a mean 0, (i.e. E[g]=0) with variance σ_A^2. The expectation of our sample mean is the mean of the larger population E[$\bar{g}$] = 0.

(i) What is the variance of the sample mean about the true mean of our infinite pool? This is the variance of a sample mean, i.e. σ_A^2/s.

(ii) What is the genetic variance <u>within</u> our base generation? This is $\sigma_A^2(1-1/s)$.

These variances can be derived from the **A** matrix, using a general method for calculating both (i) and (ii) for any group of individuals. As an example consider any set of individuals '1', '2', '3' with g_1, g_2, g_3.

$$\text{var}(\bar{g}) = \text{var}((g_1+g_2+g_3)/3)$$

$$= 1/9 \times [\text{var}(g_1) + \text{var}(g_2) + \text{var}(g_3) + 2\text{cov}(g_1,g_2) + 2\text{cov}(g_1,g_3) + 2\text{cov}(g_2,g_3)]$$

$$= 1/9 \times [\text{var}(g_1)+\text{var}(g_2)+\text{var}(g_3)+\text{cov}(g_1,g_2)+\text{cov}(g_2,g_1)+\text{cov}(g_1,g_3)+\text{cov}(g_3,g_1)+\text{cov}(g_2,g_3)+\text{cov}(g_3,g_2)]$$

... just using $\text{cov}(g_2,g_3)=\text{cov}(g_3,g_2)$

$$= \sigma_A^2/9 \times [a_{11} + a_{12} + a_{13} + a_{21} + a_{22} + a_{23} + a_{31} + a_{32} + a_{33}]$$ … by definition of **A**

This is simply the average of all the 9 pairwise numerator relationships among the 3 individuals. In the base the covariances are 0, so this simplifies to $\sigma_A^2/3$.

The variance within the group is:

$$E[\text{var}(g)] - \text{var}(\bar{g})$$

$$= \sigma_A^2 \times [a_{11} + a_{22} + a_{33}]/3 - \sigma_A^2/9 \times [a_{11} + a_{12} + a_{13} + a_{21} + a_{22} + a_{23} + a_{31} + a_{32} + a_{33}]$$

$$= \sigma_A^2 - \sigma_A^2/3$$ …. for the individuals in the base

See the box below to generalise this process for any subset. Using this process will give the results given above for a base generation of *s* individuals:

So when we say the variance σ_A^2 refers to the base generation, strictly we mean the imaginary distribution from which the base is sampled, or a base generation that is very large in size.

To generalise the process where I have **A** for a population of *n* individuals, with genetic variance σ_A^2 associated with it, and I wish to consider a subset of *s* individuals. Define **c** as a vector of 0's and 1's defining which individuals are in the subset and which are not, i.e. $c_i = 1$ if individual *i* is in the subset and 0 if it is not.

The variance of the subset mean is $\mathrm{var}(\bar{g}) = \sigma_A^2\ \mathbf{c}^T\mathbf{A}\mathbf{c} / s^2$ and the genetic variance within the subset is $(\sum_1^n c_i a_{ii} / s - \mathbf{c}^T\mathbf{A}\mathbf{c} / s^2)\ \sigma_A^2 = k\sigma_A^2$ where *k* is a coefficient derived from the known **A** matrix.

We can write *k* more concisely … it is the mean of the diagonal of the sub-matrix of **A** corresponding to our subset minus the mean of all the elements of the sub-matrix i.e. $\overline{diag(\mathbf{A}(s)} - \overline{\mathbf{A}(s)}$ where $\mathbf{A}(s)$ is the sub-matrix corresponding to the subset.

A is not special, and we can substitute a genomic matrix **G** with a genomic variance σ_g^2 to calculate the variances associated with a subset as predicted by the genomic relationship matrix: i.e. the genetic variance within the subset is $\overline{diag(\mathbf{G}(s)} - \overline{\mathbf{G}(s)}\ \sigma_g^2 = k\ \sigma_g^2$ where *k* is a coefficient derived from the known **G** matrix.

GENETIC VARIANCES CAN BE COMPARED!

It is possible to compare variance produced by the different models. Each model has a (co)variance matrix with an estimated variance, $\hat{\sigma}_A^2\mathbf{A}$, $\hat{\sigma}_g^2\mathbf{G}_1$ or $\hat{\sigma}_g^2\mathbf{G}_2$, and the genetic variance predicted for any subset can be compared using the process given above. What subset are you interested in?

What is the s.e. of the estimate of variance within the subset? The estimate is simply $k\hat{\sigma}_A^2$ or $k\hat{\sigma}_g^2$, for some coefficient *k* derived from **A** (or **G)** as described in the box above. Therefore the s.e. for the estimate of the variance within the subset is simply *k* times the s.e. of the estimate of $\hat{\sigma}_A^2$ (or $\hat{\sigma}_g^2$) given by ASReml.

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